Aakash
Medical | IIT-JEE | Foundations
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NCERT BOOSTER TEST SERIES

Test No. –06

Topics covered in various subjects:

Physics : Electric Charges and Fields, Electrostatic Potential and Capacitance

Chemistry : Electrochemistry, Chemical Kinetics

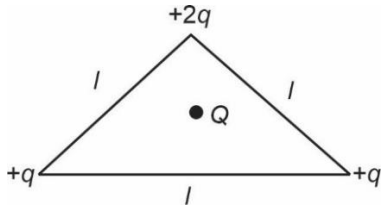
Botany : **Molecular Basis of Inheritance :** Transcription Unit and The Gene, Types of RNA and the process of Transcription, Genetic Code- Mutations and Genetic code, tRNA - the Adapter Molecule, Translation, Regulation of Gene Expression - The Lac Operon, Human Genome Project, DNA Fingerprinting, **Strategies of Enhancement in food Production**

Zoology : Evolution, Human Health and Disease, **Strategies for Enhancement of Food Production:** Animal Husbandry

PHYSICS

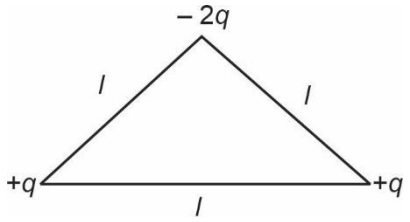
Choose the correct answer:

1. The name of apparatus to detect the charge on a body is
   (1) Potentiometer
   (2) Electron microscope
   (3) Cyclotron
   (4) Electroscope
2. Which among the followings is not a basic property of electric charge?
   (1) Additivity of charge
   (2) Conservation of charge
   (3) Multiplication of charge
   (4) Quantisation of charge
3. If 2×10^9 electrons move out of a body to another body every second. Then time required to get a total charge of 4C on the other body will be
   (1) 1.25×10^{10} s (2) 1.25×10^8 s
   (3) 2.5×10^{10} s (4) 2.5×10^8 s
4. The ratio of acceleration of electron and proton due to electrical force of their mutual attraction when they are separated by a distance 1A°, will be ($m_p = 1.67 \times 10^{-27}$ kg and $m_e = 9.11 \times 10^{-31}$ kg)
   (1) 1.8×10^3 (2) 2.5×10^2
   (3) 1.4×10^3 (4) 2.5×10^3
5. A body contains 4n proton and n electron. If the magnitude of charge of electron is e then the total amount of charge on the body will be
   (1) 3ne (2) – 3ne
   (3) 5ne (4) – 5ne
6. Three charges q, q and $2q$ are placed at three vertices of an equilateral triangle of side l. The magnitude of net force on the charge Q placed at the centroid of the triangle as shown in the figure will be



   (1) $\frac{3Qq}{4\pi\varepsilon_0 l^2}$ (2) $\frac{Qq}{4\pi\varepsilon_0 l^2}$
   (3) $\frac{3Qq}{2\pi\varepsilon_0 l^2}$ (4) Zero

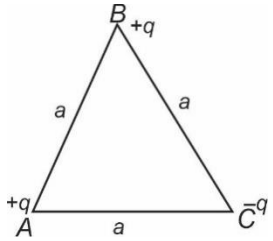
7. Consider the charges q, q and $-2q$ placed at the vertices of an equilateral triangle as shown in the figure. The net electrostatic force on the system of charges will be



(1) $\frac{\sqrt{3}q^2}{2\pi\varepsilon_0 l^2}$ (2) $\frac{\sqrt{5}q^2}{4\pi\varepsilon_0 l^2}$

(3) $\frac{q^2}{4\pi\varepsilon_0 l^2}$ (4) Zero

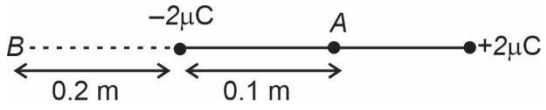
8. Consider the three particle A, B and C having charge $+q$, $+q$ and $-q$ are placed at the vertices of an equilateral triangle of side a as shown in the figure. The net electrostatic force on the particle B will be



(1) $\frac{\sqrt{3}q^2}{4\pi\varepsilon_0 a^2}$ (2) $\frac{q^2}{4\pi\varepsilon_0 a^2}$

(3) $\frac{2q^2}{4\pi\varepsilon_0 a^2}$ (4) Zero

9. Two point charges q_1 and q_2 are of magnitude + 2μC and – 2μC, respectively, are placed 0.2 m apart as shown in figure. The ratio of electric field at a point A to the point B will be



(1) $\frac{32}{3}$ (2) $\frac{4}{3}$

(3) $\frac{3}{2}$ (4) $\frac{1}{3}$

10. Choose the correct statement among the followings.

(1) Field lines start from positive charges and end at negative charges. If there is a single charge, they may start or end at infinity

(2) In a charge-free region, electric field lines can be taken to the continuous curves without breaks

(3) Two field lines can never cross each other

(4) All of these

11. **Statement(A):** The total charge of electric dipole is obviously zero. This means that the electric field due to electric dipole will be zero.

Statement(B): Direction of electric dipole moment is from negative charge to positive charge.

(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both the statements (A) and (B) are correct

(4) Neither statement (A) nor (B) is correct

12. **Statement(A):** Gauss's law is true for any closed surface, no matter what is shape and size.

Statement(B): Gauss's law is based on the inverse square dependence on distance contained in the Coulomb's law.

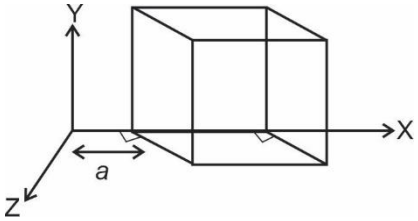
(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both the statements (A) and (B) are correct

(4) Neither statement (A) nor (B) is correct

13. The electric field components in figure $E_x = Kx$, $E_y = 0$, $E_z = 0$, The net charge within the cube of side a will be



(1) $2ka^3\varepsilon_0$ (2) $ka^3\varepsilon_0$

(3) $3ka^3\varepsilon_0$ (4) Zero

14. The dimensional formula of the electric flux is

(1) $[ML^3T^{-3}A^{-1}]$ (2) $[ML^2T^{-3}A]$

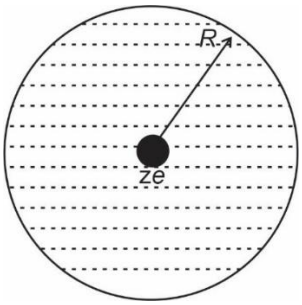
(3) $[ML^3T^{-2}A]$ (4) $[MLT^{-2}A^{-1}]$

15. The magnitude of electric field due to a spherical shell of radius R having surface charge density σ at distance $r > R$ will be

(1) $\frac{\sigma R}{\varepsilon_0 r}$ (2) $\frac{\sigma R^3}{\varepsilon_0 r^3}$

(3) $\frac{\sigma R^2}{\varepsilon_0 r^2}$ (4) $\frac{\sigma}{\varepsilon_0 R^2}$

16. An early model for an atom considered it to have a positively charged point nucleus of charge Ze, surrounded by a uniform density of negative charge up to a radius R. The atom as a whole is neutral as shown in the figure. The magnitude of electric field at a distance $r = \frac{R}{2}$ from centre will be



(1) $\frac{Ze}{8\pi\varepsilon_0 R^2}$ (2) $\frac{7Ze}{8\pi\varepsilon_0 R^2}$

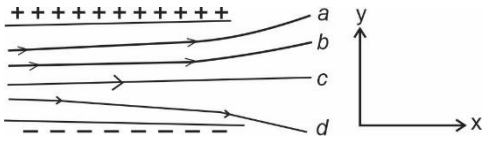
(3) $\frac{3Ze}{8\pi\varepsilon_0 R^2}$ (4) $\frac{3Ze}{4\pi\varepsilon_0 R^2}$

17. The ratio of magnitude of electric field due to an electric dipole having dipole moment P at a distance r at its equatorial position to axial position will be (where r is distance of the point from centre of the dipole and $2a$ is distance between the charges of dipole)

(1) $\frac{r\left(r^2-a^2\right)^2}{2\left(r^2+a^2\right)^{3/2}}$ (2) $\frac{\left(r^2+a^2\right)^{3/2}}{2r\left(r^2-a^2\right)^2}$

(3) $\frac{1}{2}$ (4) $\frac{\left(r^2-a^2\right)^2}{2r\left(r^2+a^2\right)^{3/2}}$

18. Figure as given below shows the tracks of four charged particles in a uniform electrostatic field. Which particle has the highest charge to mass ratio?



(1) a (2) b

(3) d (4) c

19. The electrostatic force on a small sphere(A) of charge 0.2μc due to another small sphere(B) of charge – 4μc in air is 0.4 N. The force on the sphere (B) by sphere(A) will be

(1) 0.4 N repulsive (2) 0.8 N repulsive

(3) 0.4 N attractive (4) 0.8 N attractive

20. Four point charges $q_A = 6\mu C$, $q_B = -8\mu C$, $q_C = 6\mu C$ and $q_D = 8\mu C$ are located at the corners of a square ABCD of side $3\sqrt{2}$ cm. The net electrostatic force on a 1μc placed at the centre of the square will be

(1) 80 N (2) 60 N

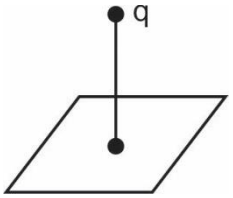
(3) 120 N (4) 160 N

21. An electric dipole of dipole moment 5×10^{-9} Cm is aligned at 37° with the direction of uniform electric field of magnitude 10^5 NC^{-1}. The magnitude of torque acting on the dipole will be

(1) 1.5×10^{-4} Nm (2) 1.5×10^{-5} Nm

(3) 3×10^{-4} Nm (4) 3×10^{-5} Nm

22. A point charge q is at a distance a directly above the centre of a square of side $2a$ as shown in figure. The electric flux through the square will be



(1) $\frac{q}{\varepsilon_0}$ (2) $\frac{q}{4\varepsilon_0}$

(3) $\frac{q}{6\varepsilon_0}$ (4) $\frac{q}{3\varepsilon_0}$

23. Two parallel infinite line charge having equal and opposite charge density separated by a distance 4 cm produce a field 18×10^4 N/C midway between them. The magnitude of linear charge density of line charges will be

(1) $0.2\ \mu Cm^{-1}$ (2) $0.1\mu Cm^{-1}$

(3) $0.15\ \mu Cm^{-1}$ (4) $0.5\mu Cm^{-1}$

24. The work done in bringing a charge of 4×10^{-9} C from infinity to a point P which is 9 cm away from a fixed-point charge of 5×10^{-7} C is

(1) 2×10^{-5} J (2) 2×10^{-4} J

(3) 2×10^{-3} J (4) 2×10^{-6} J

25. The electrostatic potential due to an electric dipole of dipole moment p at a distance r from its centre will be (where $r >> a$ and a is distance between the charges of the electric dipole)

(1) $\frac{\vec{p} \times \vec{r}}{4\pi\varepsilon_0 r^2}$ (2) $\frac{\vec{p}.\vec{r}}{4\pi\varepsilon_0 r^2}$

(3) $\frac{\vec{p}.\vec{r}}{4\pi\varepsilon_0 r^3}$ (4) $\frac{\vec{p} \times \vec{r}}{4\pi\varepsilon_0 r^3}$

26. Two charges + q and – 3q are placed at points A and B respectively separated by 40 cm. The distance of a point from point A between the charges where electric potential is zero

(1) 10 cm (2) 20 cm

(3) 30 cm (4) 25 cm

27. The equipotential surface due to the uniform electric field will be

(1) Spherical (2) Cylindrical

(3) Planar (4) Circular

28. **Statement(A):** Electric field is in the direction in which the potential decreases steepest.

Statement(B): The magnitude of electric field is given by the change in magnitude of potential per unit displacement normal to the equipotential surface at the point.

(1) Only statement (A) is correct

(2) Only statement (B) is correct

(3) Both the statements (A) and (B) are correct

(4) Neither statement (A) nor (B) is correct

29. Three charges + q, + q and – 2q are placed at the corners of an equilateral triangle of side a. The work done required to bring a charge q_0 from infinity to the centroid of the triangle will be

(1) $\frac{\sqrt{3}qq_0}{\pi\varepsilon_0 a}$ (2) $\frac{\sqrt{3}qq_0}{4\pi\varepsilon_0 a}$

(3) $\frac{qq_0}{\pi\varepsilon_0 a}$ (4) Zero

30. Two charges 6μC and – 4μC are placed 9 cm apart. The work done required to separate the two charges infinitely away from each other will be

(1) 1.2 J (2) 4.8 I

(3) 2.4 J (4) 3.6 J

31. Choose the correct statement among the following.

(1) Inside a conductor, electrostatic field is zero

(2) At the surface of charged conductor, electrostatic field must be normal to the surface every point

(3) The interior of a conductor can have no excess charge in the static condition

(4) All of these

32. Which among the following molecule is polar?

(1) O_2 (2) H_2

(3) HCl (4) Both (1) & (2)

33. Choose the correct relation between polarisation P and electric field E.

(1) $P = \frac{E}{\chi_e}$ (2) $P = \chi_e E$

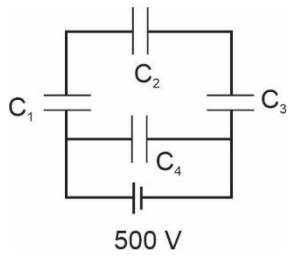
(3) $P = \varepsilon_0 (k - 1)E$ (4) Both (2) & (3)

34. A slab of material of dielectric constant K has the same area as the plates of a parallel plate capacitor of capacitance C but has a thickness $\frac{2d}{3}$, (where d is the separation of the plates), inserted between the plates of the capacitor then the new capacitance of the capacitor will be

(1) $\frac{CK}{K+2}$ (2) $\frac{2CK}{K+3}$

(3) $\frac{3CK}{K+2}$ (4) C

35. A network of four capacitor each of capacitance 30μF are connected to a 500 V supply as shown in the figure. The ratio of charge on capacitor C_2 to the capacitor C_4 will be



(1) 1: 1 (2) 1: 3

(3) 1: 2 (4) 1: 4

36. The dimensional formula of polarisation *P* of dielectric is

(1) $[L^{-1}AT]$ (2) $[L^{-2}AT]$

(3) $[L^{-2}A^{-1}T]$ (4) $[L^{2}AT^{2}]$

37. **Statement(A):** Electric field is discontinuous across the surface of a spherical charged shell.

Statement(B): Potential due to a charge *q* at its own location is not defined.

(1) Only statement(A) is correct

(2) Only statement(B) is correct

(3) Both the statements (A) and (B) are correct

(4) Neither statement (A) nor statement (B) is correct

38. A regular hexagon of side 20 cm has a charge 6μC at each of its vertices. The net electrostatic potential at the centre of the hexagon will be

(1) 5.4×10^6 V (2) 10.8×10^6 V

(3) 3.48×10^6 V (4) 1.62×10^6 V

39. Three capacitors of capacitances of 4 pF, 2 pF and 4 pF are connected in the series, the equivalent capacitance of the combinations will be

(1) $\frac{2}{3}$pF (2) 1 pF

(3) $\frac{4}{3}$pF (4) 2 pF

40. A 6 pF capacitor is connected to 25 V battery, the ratio of energy stored in the capacitor to the work done by the battery will be

(1) 1 : 1 (2) 1 : 2

(3) 1 : 4 (4) 1 :3

41. The capacitance of a parallel plate capacitor is independent of

(1) Area of the plate

(2) Charge on the plate

(3) Medium between the plate

(4) Both (1) and (3)

42. Charge density of plates of parallel plates capacitor are σ and – σ, then magnitude of electric field between the plates will be

(1) $\frac{2\sigma}{\varepsilon_0}$ (2) $\frac{\sigma}{2\varepsilon_0}$

(3) $\frac{\sigma}{\varepsilon_0}$ (4) Zero

43. A 600 pF capacitor is charged by a 200 V supply. It is then disconnected from the supply and is connected to another uncharged 600 pF capacitor, the electrostatic energy lost in this process is

(1) 3×10^{-6} J (2) 4×10^{-6} J

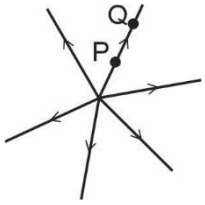
(3) 2×10^{-6} J (4) 6×10^{-6} J

44. A spherical conductor of radius 6 cm has a charge of 1.6×10^{-8} C distributed uniformly on the surface. The ratio of electric field just outside the sphere to the electric field at a point 12 cm from the centre of the sphere will be

(1) 2 : 1 (2) 4 :1

(3) 1 : 1 (4) 3 : 1

45. Figure shows the field lines of a positive charge. If a small negative charge is released at point Q then the sign of potential energy difference of negative charge between the points Q and P will be



(1) Positive

(2) Negative

(3) Zero

(4) Either positive or negative

CHEMISTRY

46. Standard electrode potential at 298 K for the reaction $2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ is

(1) 0 V (2) –0.83 V

(3) 1.7 8 V (4) 1.23 V

47. For the cell $Mg|Mg^{2+}$ (0.01M)||Ag^{+} (0.01M)|Ag, E°_{Cell} = 3.17 V. The difference between E°_{Cell} and E_{Cell} is

(1) 0 V (2) $\frac{0.591}{2}$ V

(3) 0.0591 V (4) 2 × 0.591 V

48. The correct order of conductivity of metals at 298.15 K is

(1) Au > Ag > Cu (2) Ag > Au > Cu

(3) Cu > Ag > Au (4) Ag > Cu > Au

49. The conductivity of electrolytic (ionic) solutions increases with

(1) Increase in solvation of ions

(2) Viscosity of solvent

(3) Concentration of the electrolyte

(4) Decrease in temperature

50. Cell constant is equal to

(1) Conductivity × resistance

(2) $\frac{\text{Conductivity}}{\text{Resistance}}$

(3) $\frac{l}{A}$

(4) Both (1) and (3)

51. Select the correct statement regarding conductivity on dilution

(1) It decreases for strong and weak electrolytes both

(2) It increases for strong and weak electrolytes both

(3) It increases for weak but decreases for strong electrolyte

(4) It increases for strong but decreases for weak electrolyte

52. Intercept of plot $\wedge_m$ against $C^{1/2}$ for strong electrolyte is –A. The value of A for NaCl is same as of

(1) KCl (2) $CaCl_2$

(3) $MgSO_4$ (4) $AlCl_3$

53. Maximum limiting molar conductivity among the following ions at 298 K is

(1) Na^{+} (2) K^{+}

(3) Ca^{2+} (4) Mg^{2+}

54. During the electrolysis of aqueous sodium chloride, which of the following product can not be obtained?

(1) Na (2) NaOH

(3) Cl_2 (4) H_2

55. In Leclanche cell, ammonia produced in the reaction converts into

(1) NH_4Cl (2) $[Zn(NH_3)_4]^{2+}$

(3) N_2 (4) NH_4OH

56. In mercury cell, overall reaction does not involve any ion in solution therefore

(1) Cell potential is greater than 1V

(2) Cell potential remains constant

(3) Cell potential is very tow

(4) Cell is rechargeable

57. On charging the lead storage battery, $PbSO_4$ converted into

(1) PbO (2) PbO_2

(3) Pb (4) Both (2) and (3)

58. During the corrosion of iron, the reaction taking place at cathode is

(1) $O_2(g) + 4H^{+}(aq) + 4e^{-} \rightarrow 2H_2O(l)$

(2) $2Fe^{2+}(aq) + 2H_2O(l) \xrightarrow[4H^{+}]{} 2Fe(s) + O_2(g) +$

(3) $2Fe(s) \rightarrow 2Fe^{2+}(aq) + 4e^{-}$

(4) $2H^{+}(aq) + 2e^{-} \rightarrow H_2(g)$

59. Which of the following is true for a galvanic cell at equilibrium

(1) $\Delta G = 0$ (2) K = 0

(3) $\Delta G^{\circ} = 0$ (4) All of these

60. In order to prevent the corrosion of iron, which metal can't be used?

(1) Mg (2) Zn

(3) Sn (4) Cu

61. The amounts of different substances liberated by the same quantity of electricity passing through the electrolytes are in the ratio of their

(1) Molar mass (2) Atomic number

(3) Equivalent mass (4) Valency

62. If Λ^{0}_{m} for HA_C, NaCl and HCl are 390.5, 126.4 and 425.9 $Scm^2\ mol^{-1}$ respectively then Λ^{0}_{m} NaA_C is (in Scm^2mol^{-1})

(1) 91 (2)109

(3) 211 (4) 322

63. Λ_m^0 (KCl) – Λ_m^0 (NaCl) is equal to

(1) Λ_m^0 (KCl) – Λ_m^0 (KBr)

(2) Λ_m^0 (KBr) – Λ_m^0 (NaBr)

(3) Λ_m^0 (KCl) – Λ_m^0 (NaBr)

(4) Λ_m^0 (NaBr) – Λ_m^0 (KBr)

64. During the electrolysis of aqueous solution of $AgNO_3$ using sliver electrode, which of the following statement is/are true

(1) Ag(s) gets converted into Ag^+(aq) at anode

(2) Ag^+(aq) gets converted into Ag(s) at cathode

(3) Reduction of NO_3^- takes place at cathode

(4) Both (1) & (2)

65. Faraday of electricity required to produce 80 g of calcium from molten $CaCl_2$ is

(1) 1 F (2) 2 F

(3) 4 F (4) 8 F

66. Greater is the reduction potential, lesser is

(1) Reducing nature (2) Reducing power

(3) Reducing character (4) All of these

67. SHE is

(1) Pt(s) | H_2(g, 1bar) | H^+(aq, 1M)

(2) Pt(s) | H_2(g, 1atm) | H^+(aq, 1M)

(3) Pt(s) | H_2(g, 1bar) | H^+(aq, 0.1M)

(4) Pt(s) | H_2(g, 1atm) | H^+(aq, 0.1M)

68. Example of very fast reaction is

(1) Precipitation of silver chloride

(2) Rusting of iron

(3) Hydrolysis of starch

(4) Inversion of cane sugar

69. For the reaction

$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \rightarrow 3Br_2(aq) + 3H_2O(l)$ rate at reaction is/are

(1) $\frac{1}{5}$ times rate of disappearance of Br^-(aq)

(2) 6 times rate of disappearance of H^+(aq)

(3) –1 times rate of disappearance of BrO_3^- (aq)

(4) All of these

70. For the reaction $2A \rightarrow B$, the concentration of reactant A changes from 0.04 M to 0.02 M in 2 min. The average rate of reaction is

(1) 0.01 Ms^{-1} (2) 1.66×100^{-4} Ms^{-s}

(3) 8.33×10^{-5} Ms^{-1} (4) 5×10^{-3} Ms^{-1}

71. For the reaction $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$. On doubling the concentration of NO while keeping the concentration of O_2 constant. The initial rate increases by a factor of four. The order with respect NO(g) is

(1) 0 (2) 1

(3) 2 (4) 3

72. Which of the following rate expression is/are possible?

(1) Rate = $k[A]^{1/2}[B]^{3/2}$

(2) Rate = $k[A]^{3/2}[B]^{-1}$

(3) Rate = $k\,[A]^1[B]^2$

(4) All of these

73. Unit of rate constant for second order reaction is?

(1) s^{-1} (2) mol L^{-1}

(3) mol $L^{-1}s^{-1}$ (4) $mol^{-1}Ls^{-1}$

74. Which of the following statement is incorrect?

(1) Termolecular reactions involve simultaneous collision between three reacting species

(2) The probability that more than three molecules can collide is very small

(3) Reaction with the molecularity three are very common

(4) Reaction with molecularity three are slow to proceed

75. The slope of the graph between the concentration vs time for a zero order of reactant reaction is

(1) k (2) –k

(3) E_a (4) $-E_a$

76. Fraction of concentration of reactant left after time t for a first order reaction is

(1) e^{-kt} (2) –kt

(3) $e^{-k/t}$ (4) $e^{-t/k}$

77. For a first order reaction

$2N_2O_5(g) \rightarrow 2N_2O_4(g) + O_2(g)$. If initial pressure of N_2O_5 is 2 atm then total pressure at $t = t_{1/2}$ is

(1) 1 atm (2) $\frac{5}{2}$ atm

(3) 3 atm (4) 6 atm

78. For a first order reaction, $t_{1/2}$ is proportional to

(1) $[R]_0^0$ (2) $[R]_0^1$

(3) $[R]_0^{-1}$ (4) $[R]_0^{-2}$

79. Ratio of time required for 99% completion vs 90% completion of a first order reaction is

(1) $\frac{99}{90}$ (2) $\frac{9}{10}$

(3) $\frac{10}{9}$ (4) $\frac{2}{1}$

80. Inversion of cane sugar is an example of

(1) Zero order reaction

(2) First order reaction

(3) Pseudo first order reaction

(4) Pseudo second order reaction

81. Addition of catalyst

(1) Catalyses the forward reaction only

(2) Catalyses the backward reaction only

(3) Catalyses the forward as well as backward reactions to the same extent

(4) Catalyses the forward as well as backward reactions to the different extent

82. Select the correct expression on the basis of collision theory

(1) Rate = $PZ_{AB}\, e^{-Ea/RT}$

(2) Rate = $A\, e^{-Ea/RT}$

(3) Rate = $PZ_{AB}\, e^{Ea/RT}$

(4) Rate = $Ae^{Ea/RT}$

83. If for a reaction, values of rate constants at 10 K and 20K are 0.1 s^{-1} and 0.2 s^{-1} respectively. The activation energy of the reaction is

(1) 152.2 KJ mol^{-1} (2) 115.2 J mol^{-1}

(3) 57.63 J mol^{-1} (4) 57.63 KJ mol^{-1}

84. $-\frac{Ea}{R}$ is slope of the plot between

(1) ln k vs T (2) ln k vs $\frac{1}{T}$

(3) log k vs T (4) log k vs $\frac{1}{T}$

85. Select the correct statement

(1) Order is applicable to elementary as well as complex reactions

(2) Order of a reaction is an experimental quantity

(3) Molecularity is not valid for complex reaction

(4) All of these

86. The decomposition of gaseous ammonia on a hot platinum surface is _____ order reaction at high pressure

(1) 0 (2) 1

(3) 2 (4) 3

87. The half life of a first order reaction if rate constant is 100 s^{-1} is

(1) 693 s (2) 69.3 s

(3) 0.693 s (4) 6.93×10^{-3} s

88. 1 faraday is the quantity of electricity which produces

(1) 1 g of substance

(2) 1 mole of substance

(3) 1 equivalent of substance

(4) 22.4 g of substance

89. Device used to convert energy of combustion directly into electrical energy is known as

(1) Galvanic cell (2) Fuel cell

(3) Electrolytic cell (4) Mercury cell

90. Which reaction can go for completion?

(1) 0 order (2) 1st order

(3) 2nd order (4) 3rd order

BOTANY

91. The functional unit of inheritance is

(1) mRNA (2) Gene

(3) DNA (4) Polypeptide

92. DNA sequence can code for all of the following, **except**

(1) dAMP (2) rRNA

(3) mRNA (4) tRNA

93. Monocistronic structural genes are found mostly in

(1) Bacteria (2) Blue-green algae
(3) Monerans (4) Eukaryotes

94. Select the **incorrect** statement regarding exons
(1) These are absent in processed RNA
(2) These are coding sequences
(3) These are interrupted by introns
(4) Monocistronic structural genes have exons

95. In organisms, the inheritance of characters is **not** affected by
(1) Regulatory sequence
(2) Promoter sequence
(3) Amino acid sequence in polypeptide
(4) Structural gene sequence

96. Which of the following RNAs plays both structural and catalytic role during translation?
(1) mRNA (2) rRNA
(3) tRNA (4) hnRNA

97. To catalyse the transcription of all types of RNA in bacteria, how many type(s) of DNA-dependent RNA polymerase is/are present in them?
(1) Two (2) Three
(3) Five (4) One

98. Which enzyme in bacteria facilitates opening of the DNA helix and continues elongation for transcription?
(1) DNA helicase (2) Topoisomerase
(3) Ribozyme (4) RNA polymerase

99. The RNA polymerase(s) that transcribe(s) precursor of mRNA is/are
(1) RNA polymerase I and III
(2) RNA polymerase II
(3) RNA polymerase II and III
(4) RNA polymerase III

100. The first step of processing hnRNA is
(1) Addition of adenylate residues at the 3' end
(2) Addition of methyl guanosine triphosphate to the 5' end
(3) Removal of introns from hnRNA
(4) Joining of axons in a defined order

101. Which of the following represents probably an ancient feature of the genome?
(1) Split-gene arrangements
(2) Clear-cut division of labour in transcription
(3) Involvement of RNA polymerase during transcription
(4) Transcription of all types of RNAs from DNA

102. The existance of complementarity is not seen during the process of
(1) Replication (2) Reverse transcription
(3) Translation (4) Transcription

103. The scientist who first suggested that in order to code for all the 20 amino acids, the code should be made up of three nucleotides was a
(1) Physicist (2) Botanist
(3) Geneticist (4) Biochemist

104. Which one of the following enzymes is helpful in polymerising RNA with defined sequences in a template independent manner?
(1) Kornberg enzyme
(2) Severo Ochoa enzyme
(3) Topoisomerase
(4) Ligase

105. The first base towards 5' end of mRNA for all the stop codons of universal genetic code is
(1) Guanine (2) Adenine
(3) Cytosine (4) Uracil

106. The genetic codon that has dual function, codes for (i) and also acts as (ii) codon.
(1) (i) – Phenylalanine; (ii) start
(2) (i) – Methionine; (ii) Initiator
(3) (i) – Alanine; (ii) Start
(4) (i) – Methionine; (ii) Terminator

107. A classical example of point mutation that results into a diseased condition is
(1) Sickle cell anaemia
(2) Leukemia
(3) Down's syndrome
(4) Turner's syndrome

108. Transfer RNA was formerly referred to as
(1) Transporter RNA (2) Soluble RNA
(3) Carrier RNA (4) Translator RNA

109. For which of the following codons there is no tRNA

(1) GUA (2) AUG

(3) UGG (4) UAG

110. In actual structure, the tRNA is a compact molecule which looks like

(1) Tetrahedron (2) Clover-leaf

(3) Inverted L (4) Inverted T

111. Read the following statements and choose the option which is true for them

Statement A: In the first phase of translation, there is aminocylation of tRNA.

Statement B: Peptide bond formation does not require energy but the rate is enhanced in the presence of catalyst.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

112. In an mRNA, UTRs are present

(1) Just after start codon

(2) Just before stop codon

(3) Only at 5'-end

(4) At both 5'-end and at 3'-end

113. In eukaryotes, the regulation of gene expression can be exerted at all of the following levels, **except**

(1) Chromatids separation level

(2) Post transcriptional processing level

(3) Translational level

(4) Transcriptional level

114. Match the following columns w.r.t. *lac* operon and select the **correct** option.

	Column I	**Column II**
a	Regulatory gene	(i) Repressor protein binding site
b	Operator region	(ii) Inducer gene
c	*z* gene	(iii) Codes for permease
d	*y* gene	(iv) Codes for β-gal

(1) a (i), b (iii), c (iv), d (ii)

(2) a (ii), b (iv), c (i), d (iii)

(3) a (ii), b (i), c (iv), d (iii)

(4) a (i), b (iv), c (iii), d (ii)

115. Which gene in *lac* operon synthesizes mRNA all the time, *i.e.* constitutively?

(1) *z* gene (2) *a* gene

(3) *i* gene (4) *y* gene

116. The project of sequencing human genome was launched in (a) and completed in (b) .

Select the **correct** option to fill in the blanks (a) and (b).

	(a)	(b)
(1)	1990	2003
(2)	2000	2013
(3)	1983	2000
(4)	1985	1998

117. The methods involved in sequencing, human genome involved two major approaches which are

(1) Genes fragmentation and their rearrangement

(2) Genes sequencing and their multiplication

(3) Hybridisation and polymerisation of DNA

(4) Expressed sequence tags and sequence annotation

118. The sequence of which human chromosome was sequenced in the last?

(1) Y chromosome (2) X chromosome

(3) Chromosome 22 (4) Chromosome 1

119. Which of the following statements w.r.t. features of human genome is/are **not** true?

(i) The average gene consists of 30000 bases.

(ii) Less than 2 percent of the genome codes for proteins.

(iii) Chromosome X has most genes and the Y has the fewest genes.

(iv) There are about 1.4 million locations where SNPs occur.

(1) (i), (iii) & (iv) (2) (iii) only

(3) (ii) & (iv) (4) (i) & (iii) only

120. Polymorphism in DNA sequence arises due to

(1) Fragmentation (2) Mutation

(3) DNA replication (4) Evolution

121. What is the root of any breeding programme?

(1) Evaluation of parents

(2) Genetic variability

(3) Germplasm collection

(4) Selection of parents

122. For breeding a new genetic variety of a crop, which of the following steps is time consuming and tedious?

(1) Selection of parents

(2) Cross hybridisation among the selected parents

(3) Collection of variability

(4) Testing of superior recombinants

123. Which process is crucial to the success of the breeding objective and requires careful scientific evaluation?

(1) Selection of superior recombinants

(2) Release of new cultivars

(3) Commercialisation of new cultivars

(4) Entire collection of diverse alleles

124. Green revolution is concerned with

(1) Reforestation of cut down forests

(2) Plantation of more and more trees in urban areas

(3) Increase in food production by development of crop varieties

(4) Use of forest products in various ways

125. During the period 1960 to 2000, wheat production in India increased by about

(1) 30 million tonnes (2) 64 million tonnes

(3) 35 million tonnes (4) 89.5 million tonnes

126. Regarding enhancement in food production, Norman E. Borlaug is known for

(1) Developing the technique to produce SCP

(2) Enhancing the sugar content in sugarcane

(3) Developing semi-dwarf variety of wheat

(4) Improving the protein quality in pea

127. Which feature of *Saccharum barberi* is desirable for developing improved quality of sugarcane?

(1) Thicker stem

(2) High sugar content

(3) Capable to grow well in North India

(4) Dwarf stem

128. One of the bacterial diseases in crop plants is

(1) Black rot of crucifers

(2) Turnip mosaic

(3) Red rot of sugarcane

(4) Late bight of potato

129. A variety of mustard that is developed for making it resistance against white rust is

(1) Pusa snowball K-1 (2) Pusa Shubhra

(3) Pusa Komal (4) Pusa Swarnim

130. Pusa Sadabahar is a variety of

(1) Cauliflower (2) Cowpea

(3) Chilli (4) Gram

131. In mung bean, resistance to powdery mildew is induced by

(1) Mutations (2) Vernalisation

(3) Genetic engineering (4) Protoplast fusion

132. In somatic hybridisation

(1) Nucleus of one plant variety is introduced in the somatic cell of other plant variety

(2) Nuclei of two somatic cells of different plant varieties are fused

(3) DNA segment of required feature is incorporated in the cytoplasmic DNA of other plant

(4) Isolated protoplasts from two different varieties of plants are fused

133. The capacity to generate a whole plant from any cell/explant is called

(1) Regeneracy (2) Totipotency

(3) Differentiation (4) Impotency

134. The sources of single cell protein can be

(1) Prokaryotes only

(2) Unicellular only

(3) Prokaryotes and eukaryotes both

(4) Both unicellular and multicellular except autotrophs

135. The wheat variety which has high protein content and also been used as a donor for improving cultivated wheat is

(1) Himgiri (2) Atlas 66

(3) Sonalika (4) Kalyan Sona

ZOOLOGY

136. Evolutionary biology is the study of

(1) Geological history of earth

(2) History of life forms on earth

(3) Origin of only first form of life on earth

(4) Evolution of only diverse organic molecules on earth

137. Read the following statements

(1) Stellar distances are measured in light years

(2) When we see stars, we apparently are pepping into the past

(3) Universe is 20 million years old

(4) Big Bang theory explains origin of universe

How many of the above statements are **incorrect**?

(1) Zero (2) One

(3) Two (4) Three

138. Choose the **mismatch** w.r.t. origin

(1) Origin of earth – 4.5 bya

(2) Cellular form of life – 2 mya

(3) Non-cellular form of life – 3 bya

(4) Invertebrates were formed and active – 500 mya

139. _____ believe that life came on earth from outside in the form of spores. Choose the option which **correctly** fills the blank.

(1) Louis Pasteur

(2) Early Greek thinkers

(3) Oparin

(4) Haldane

140. Theory of spontaneous generation states that

(1) Life came out of decaying and rotting matter

(2) Life comes only from pre-existing life

(3) Formation of Life was preceded by chemical evolution

(4) All living organisms that we see today were created as such

141. In _I__, S.L. Miller, an American scientist created electric discharge in a closed flask containing _II_ at _III_. Choose the option which **correctly** fill the blanks

	I	II	III
(1)	1953	CH_4, NH_3, H_2, O_2	800°C
(2)	1935	CH_4, NH_3, H_2, H_2O	750°C
(3)	1973	CH_4, NH_3, H_2, O_2	800°C
(4)	1953	CH_4, NH_3, H_2, H_2O	800°C

142. Read the following statements. Select the option with **correct** statements only.

(a) According to Darwin, fitness means only reproductive fitness

(b) Alfred Wallace worked in Malay Archipelago island

(c) Geological history of earth closely correlates with biological history of earth

(d) Alfred Wallace had also come to similar conclusion as that of Darwin but many years later

(1) a, b, c, d

(2) a, b and c only

(3) d only

(4) a and d only

143. Embryological support for evolution proposed by Ernst Haeckel was disproved by

(1) Karl Von Frisch

(2) Karl Ernst Von Baer

(3) Ernst Karl Abbe

(4) Carl Linnaeus

144. Homologous organs are result of

(1) Divergent evolution

(2) Convergent evolution

(3) Adaptive convergence

(4) Co-evolution

145. Which of the following is an example of analogous organs?

(1) Forelimbs of all mammals

(2) Vertebrate hearts

(3) Flippers of Penguins and Dolphins

(4) Thorn of *Bougainvillea* and tendril of *Cucurbita*

146. Choose the **odd** one w.r.t. molecular homology

(1) Indicate common ancestry

(2) Similarities in proteins

(3) Similarities in genes

(4) Does not include biochemical similarities

147. Which of the following is **incorrect** w.r.t. selection of antibiotic resistant microbes?

(1) Example of natural selection

(2) Example of evolution by anthropogenic action

(3) Example of artificial selection

(4) Indicates evolution is a stochastic process

148. Which of the following pair is an example of convergent evolution?

(1) Koala, Bandicoot

(2) Tasmanian wolf, Tiger cat

(3) Wolf, Bobcat

(4) Numbat, Anteater

149. There are two species **A** and **B**. Life span of **A** is less than **B**. Based on this, which of the following is most likely to occur

(1) New life form of B appear early

(2) New life forms of A appear early

(3) Rate of appearance of new life forms does not depend on life cycle

(4) New life forms of A and B appear simultaneously

150. Novelty and brilliant insight of Darwin was

(1) Adaptive ability is inherited

(2) Resources are limited

(3) Population grow exponentially

(4) Nature keeps control over population size

151. The frequency of AA individuals in a population is

(1) Product of the probabilities of allele A on both the chromosomes

(2) Sum of the probabilities of allele A on both the chromosomes

(3) Probability of allele A on one chromosome

(4) Two times the product of the probabilities of allele A on both the chromosomes

152. Choose the **correct** match w.r.t. human evolution.

(1) *Ramapithecus* – More ape like

(2) *Homo erectus* – Did not eat meat

(3) Neanderthal man – Lived in near east and central Africa

(4) *Homo sapiens* – Arose in Africa

153. 'Good humor' hypothesis of health was disproved by

(1) Hippocrates

(2) William Harvey

(3) Early Greek Thinkers

(4) Charaka

154. The major cause of death among non-infectious disease is

(1) Cancer (2) AIDS

(3) Pneumonia (4) Hepatitis-B

155. Match the column-I with column-II and choose the **correct** option

Column-I	**Column-II**
(a) Common cold	(i) *Trichophyton*
(b) Pneumonia	(ii) *Wuchereria*
(c) Ring worm	(iii) Rhino viruses
(d) Filariasis	(iv) *Streptococcus*

(1) a-iv, b-iii,c-i,d-ii

(2) a-iii, b-iv, c-i, d-ii

(3) a-i, b-ii, c-iii, d-iv

(4) a-ii, b-i, c-iii, d-iv

156. I – High fever (39° to 40°C)

II – Stomach pain

III – Constipation

IV – Loss of appetite

These are common symptoms of which of the following disease?

(1) Tetanus (2) Filariasis

(3) Typhoid (4) Ringworm

157. Infectious agent of malaria is

(1) Female *Anopheles* mosquito

(2) Female *Culex* mosquito

(3) *Plasmodium vivax*

(4) *Aedes* mosquito

158. Choose the **mismatch** w.r.t. innate immunity.

(1) Physical barrier – Mucus

(2) Physiological barrier – Saliva

(3) Cellular barrier – Natural Killer cell

(4) Cytokine barrier – PMNL

159. Antigen binding sites on an antibody molecule are present at

(1) N-terminal of antibody

(2) C-terminal of antibody

(3) Constant region of light chains only

(4) Variable region of light chains only

160. Antitoxins provide which type of immunity?

(1) Natural active acquired immunity

(2) Artificial passive acquired immunity

(3) Natural passive acquired immunity

(4) Artificial active acquired immunity

161. Which of the following is **not** true for hepatitis-B vaccine produced by recombinant DNA technology?

(1) Contains antigenic polypeptides

(2) Large scale production of vaccine is possible by this approach

(3) Produced from killed yeast

(4) Provide active immunity

162. Choose the **correct** match w.r.t. lymphoid organs.

(1) Thymus – Secondary lymphoid organ

(2) Bone marrow – Lymphocytes become antigen sensitive

(3) Peyer's patches – Maturation of lymphocytes

(4) Primary lymphoid organ – Lymphocytes interact with antigen

163. Read the following statements and choose the **correct** option

Statement (A): In HIV, DNA of virus replicates to form more viral DNA with the help of enzyme reverse transcriptase

Statement(B): First case of AIDS was reported in the year 1891

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both A and B statements are incorrect

(4) Both A and B statements are correct

164. Oncogenic viruses have genes called

(1) Cellular oncogenes

(2) C-onc

(3) Proto-oncogenes

(4) Viral oncogenes

165. Radiation therapy is used in cancer treatment because

(1) Cancerous cells are rapidly dividing cells

(2) Radiations kill only cancerous cells

(3) Cancerous cells show metastasis

(4) Cancers cells show contact inhibition

166. Choose the **mismatch**

(1) Opioids – Leaves of *Papaver somniferum*

(2) Cannabinoids – Inflorescence of *Cannabis sativa*

(3) Coca alkaloid – *Erythroxylum coca*

(4) Morphine – Latex of Poppy plant

167. How many of the diseases given below in the box are associated with cigarette smoking?

Emphysema, coronary heart disease, gastric ulcer, cancer of urinary bladder

(1) 1 (2) 2

(3) 3 (4) 4

168. Read the following statements and choose the option with **correct** statements only

(a) Hepatitis-B is a chronic infection

(b) Addiction is a psychological attachment to euphoria associated with drugs

(c) NACO, educate people about AIDS and hepatis-B

(d) Prolonged use of anabolic steroids may result in stunted growth

(1) a, b and d only

(2) a, c and d only

(3) a, b and c only

(4) a, b, c, d

169. Select the **correct** sequence of HIV infection.

(a) Viral DNA incorporates into host genome

(b) New viral RNA is produced by the infected cell

(c) Viral RNA is introduced into cell

(d) Viral DNA is produced by reverse transcriptase

(1) b → d → a → c

(2) c → a → d → b

(3) c → d → a → b

(4) d → a → b → c

170. Fisheries include rearing, catching and selling of

(a) Fishes (b) Molluscs

(c) Crustaceans (d) Shell-fish

(1) a only (2) a and b only

(3) a and d only (4) a, b, c, d

171. Select the **incorrect** statement

(1) 70% of the word livestock population is in India

(2) Milk yield is primarily dependent on the quality of breeds

(3) Inbreeding depression can be overcome by mating with unrelated superior animal of the same breed

(4) *Hisardale* is a cross-breed

172. Select the **correct** sequence w.r.t. MOET in cow

(a) Cow produces 6-8 eggs

(b) Fertilised eggs at 8-32 cells stages, are recovered non-surgically

(c) Genetic mother is available for another round of super ovulation

(d) Cow is induced for super-ovulation

(1) a → d → b → c

(2) c → a → b → d

(3) d → a → b → c

(4) a → b → c → d

173. Bees are pollinators of all the following crop plants **except**

(1) Apple (2) Pear

(3) Wheat (4) Sunflower

174. Choose the **odd** one w.r.t. habitat.

(1) *Hilsa* (2) Sardines

(3) Mackerel (4) Rohu

175. MOET technique has been demonstrated for how many of the following animals?

Cattle, Sheep, Rabbit, Mare, Buffalo

(1) 2 (2) 3

(3) 5 (4) 4

176. 'Blue revolution' is

(1) Increase in egg production

(2) Increase in fisheries

(3) Increase in milk production

(4) Increase in wheat and rice production

177. Mating within the same breed, but having no common ancestors on either side of their pedigree up to 4-6 generations is called

(1) Inbreeding

(2) Out-crossing

(3) Cross-breeding

(4) Interspecific hybridisation

178. Select the **correct** statement w.r.t. artificial insemination.

(1) Success rate of artificial insemination is fairly low

(2) In AI, only frozen semen is used

(3) AI is not a controlled breeding method

(4) AI is a method of herd improvement in very less time

179. Aquaculture includes production of

(a) Aquatic plants (useful)

(b) Marine fish

(c) Shell-fish

(d) Fresh water fish

(1) a only (2) a and d only

(3) a, b and c only (4) a, b, c, d

180. Which immune response is involved in graft rejection?

(1) Cell mediated immune response

(2) Humoral immune response

(3) Antibodies mediated immune response

(4) B-lymphocytes mediated immune response

❑❑❑

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Regd. Office :Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

NCERT BOOSTER TEST SERIES

Test – 06 (Answer Key)

1.	(4)	39.	(2)	76.	(1)	109.	(4)	144.	(1)
2.	(3)	40.	(2)	77.	(2)	110.	(3)	145.	(3)
3.	(1)	41.	(2)	78.	(1)	111.	(3)	146.	(4)
4.	(1)	42.	(3)	79	(4)	112.	(4)	147.	(3)
5.	(1)	43.	(4)	80.	(3)	113.	(1)	148.	(4)
6.	(1)	44.	(2)	81.	(3)	114.	(3)	149.	(2)
7.	(4)	45.	(1)	82.	(1)	115.	(3)	150.	(1)
8.	(2)	46.	(2)	83.	(2)	116.	(1)	151.	(1)
9.	(1)	47.	(3)	84.	(2)	117.	(4)	152.	(4)
10.	(4)	48.	(4)	85.	(4)	118.	(4)	153.	(2)
11.	(2)	49.	(3)	86.	(1)	119.	(4)	154.	(1)
12.	(3)	50.	(4)	87.	(4)	120.	(2)	155.	(2)
13.	(2)	51.	(1)	88.	(3)	121.	(2)	156.	(3)
14.	(1)	52.	(1)	89.	(2)	122.	(2)	157.	(3)
15.	(3)	53.	(3)	90.	(1)	123.	(1)	158.	(4)
16.	(2)	54.	(1)	91.	(2)	124.	(3)	159.	(1)
17.	(4)	55.	(2)	92.	(1)	125.	(2)	160.	(2)
18.	(3)	56.	(2)	93.	(4)	126.	(3)	161.	(3)
19.	(3)	57.	(4)	94.	(1)	127.	(3)	162.	(2)
20.	(4)	58.	(1)	95.	(3)	128.	(1)	163.	(3)
21.	(3)	59.	(1)	96.	(2)	129.	(4)	164.	(4)
22.	(3)	60.	(4)	97.	(4)	130.	(3)	165.	(1)
23.	(2)	61.	(3)	98.	(4)	131.	(1)	166.	(1)
24.	(2)	62.	(1)	99.	(2)	132.	(4)	167.	(4)
25.	(3)	63.	(2)	100.	(2)	133.	(2)	168.	(1)
26.	(1)	64.	(4)	101.	(1)	134.	(3)	169.	(3)
27.	(3)	65.	(3)	102.	(3)	135.	(2)	170.	(4)
28.	(3)	66.	(4)	103.	(1)	136.	(2)	171.	(1)
29.	(4)	67.	(1)	104.	(2)	137.	(2)	172.	(3)
30.	(3)	68.	(1)	105.	(4)	138.	(2)	173.	(3)
31.	(4)	69.	(1)	106.	(2)	139.	(2)	174.	(4)
32.	(3)	70.	(3)	107.	(1)	140.	(1)	175.	(3)
33.	(3)	71.	(3)	108.	(2)	141.	(4)	176.	(2)
34.	(3)	72.	(4)			142.	(2)	177.	(2)
35.	(2)	73.	(4)			143.	(2)	178.	(1)
36.	(2)	74.	(3)					179.	(4)
37.	(3)	75.	(2)					180.	(1)
38.	(4)								

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